

Yutaro Kiyota

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PROJECTS

Embedded Spatial Audio Calibration System - C, Zephyr RTOS, Python, Raspberry Pi Pico W

- Built an embedded audio calibration system using Zephyr RTOS on Raspberry Pi Pico W, integrating an I2C OLED display, VL53L0X ToF sensor, ADC, GPIO, and USB CDC ACM communication.
- Developed a JSON based serial communication protocol between embedded firmware and a Python desktop application for device configuration, speaker registration, and calibration data logging.
- Implemented speaker latency measurement and synchronization using Python, NumPy, and chirp-based signal processing to calculate per-speaker delay compensation for multi-speaker audio alignment.

Stock Prediction Platform - Python, TypeScript, FastAPI, SQL, React, XGBoost

- Built an end-to-end full-stack stock prediction system leveraging FinBERT (NLP) for financial news sentiment analysis and XGBoost regression to predict next-day stock price movement using time-series and sentiment-based features
- Designed and implemented a relational SQL database schema to store historical stock prices, news sentiment, and model outputs, ensuring data integrity, efficient indexing, and low-latency queries
- Developed a REST API using FastAPI to serve real-time prediction and confidence metrics and integrated it with a TypeScript/React frontend for interactive data visualization

Socket Programming Project - C++, TCP/IP, Networking

- Developed a TCP client-server communication system in C++ using a custom packet framing protocol for reliable message transmission.
- Implemented a modular client-server architecture, abstracting socket initialization, connection handling, and data exchange into reusable, maintainable classes

- Implemented buffering, serialization, and error handling to improve communication reliability.

EXPERIENCE

Missouri University of Science & Technology - Satellite Research Team

Software Developer | Aug 2023 - Present

- Designed and extended an Operating System Abstraction Layer (OSAL) over FreeRTOS to standardize task scheduling and file system interfaces for CubeSat flight software.
- Designed and debugged a multithreaded Ethernet communication module, supporting concurrent data transfer between satellite subsystems

Missouri University of Science & Technology - Residential Life

Senior Resident Assistant | Aug 2024 – May 2026

- Supported residents facing academic and mental health challenges by providing resources, mentorship, and empathetic communication
- Organized community-building programs to improve resident engagement and foster an inclusive community.
- Mentored and trained first-year resident assistants, fostering leadership, collaboration, and conflict-resolution skills by introducing individually tailored approaches

SKILLS

C++ Python C Zephyr RTOS Git

UART I2C SQL FastAPI React

EDUCATION

Bachelor of Science in Computer Science

Aug 2023 -Present

Missouri University of Science & Technology

GPA: 3.97/4.0 | Dean's List